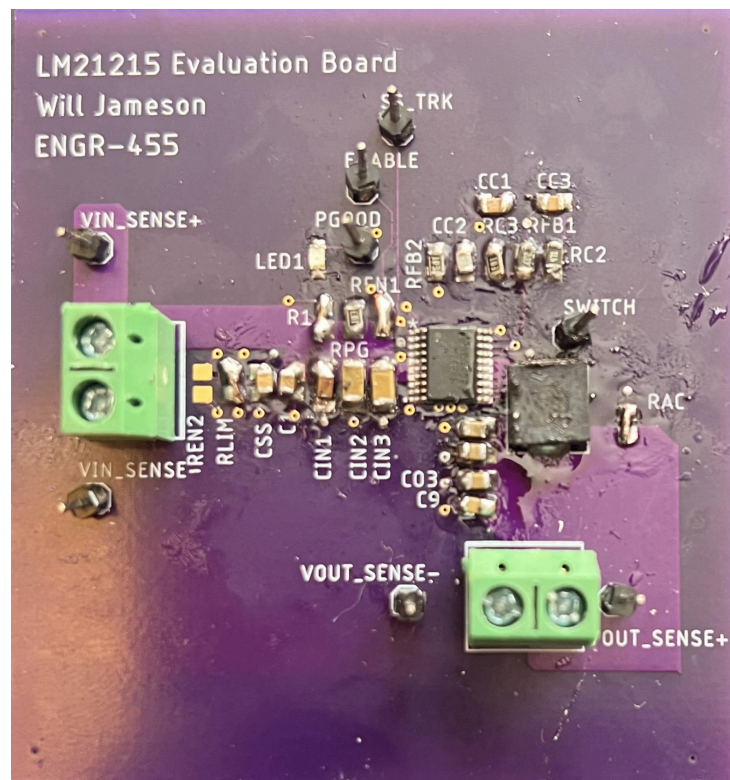


Evaluation Board for Texas Instruments LM-21215 Buck Converter

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ENGR-455

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Abstract:

A buck converter using a Texas Instruments LM21215 IC was to be designed for use as a 5V to 1.8V step down DC-DC applications. It was designed to supply 5A of current with a peak to peak inductor ripple of less than 30%. In addition, input voltage ripple and output voltage ripple were to be held to less than 1% at the input and output. The compensation loop was to have a phase margin of greater than 50 degrees, and respond to a 2 to 3 amp load step with a transient droop of no more than 50mV. Power path and compensation loop components were to be selected to adhere to these parameters.

The resulting prototype is the TI-LM21215 Evaluation board. It features test points at the input and output, as well as the switch node, P-good and enable nodes. All components were hand-soldered using 0805 or larger sized components. The following document will analyze the functionality of the board, test compliance with design parameters, and analyze unexpected behavior in the system.

Equipment Used:

- Keysight EDU36311A Benchtop Power Supply
- Keysight EDUX1052A Oscilloscope
- Keysight EDU34450A Digital Multimeter
- Keysight Electronic Load Device

Part 1: Current Ripple Test:

The inductor chosen for this circuit is a Vishay-Dale 2.2 μH high current inductor with an ESR value of $\approx 25\text{m}\Omega$. To test the current ripple, the circuit is broken and a wire loop was inserted at the V_{out} side of the inductor. Using a clamp sensor connected to the oscilloscope, the current ripple was measured.

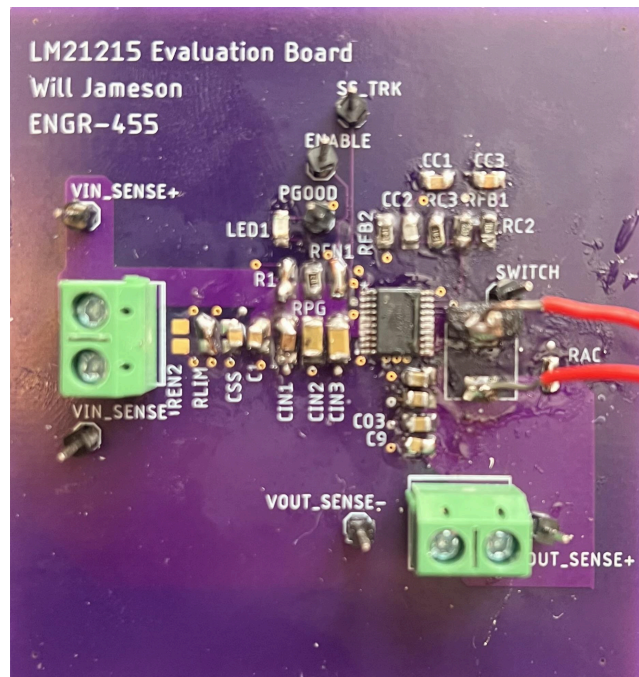


Figure 1: Red wire between inductor and output voltage node

5 Volts was applied at the input using the DC power supply, and the current ripple was viewed on the oscilloscope.

Table 1: Current Ripple Responding to Load Variation

Load (A)	Output Voltage(V)	Current Ripple _{p-p} (A)
1	1.8016	0.80625
2	1.8014	0.80625
3	1.8013	0.71875
4	1.7989	0.01580
5	1.7824	0.01490

Data Analysis:

From Table 1, we can see the ripple current collapsing before we reach the rated load. At smaller current loads, the current ripple stays robust and has a characteristic triangular waveform with a steady Δi .

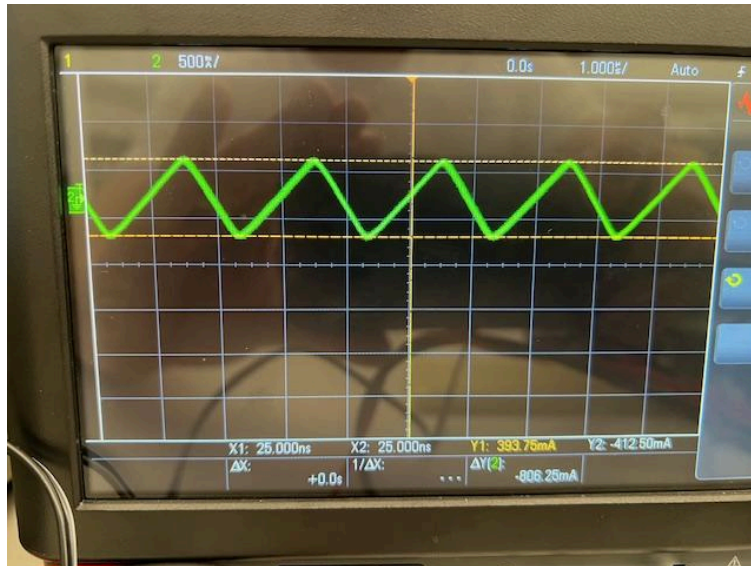


Figure 1. Inductor ripple at 2A load

The inductor ripple begins to take on a 'scooped' shape at about 3A indicating inductor saturation. At this point, the converter is still supplying the 1.8V output demanded by the design.

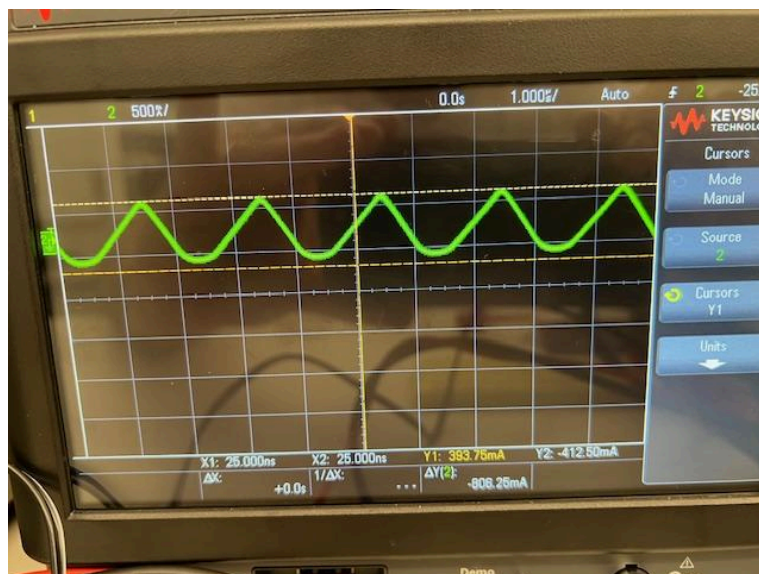


Figure 2: Scooping indicating inductor is saturating

This trend continues until the inductor current ripple begins to show more pronounced signs of saturation. The ripple collapses to nearly nothing, but even though the inductor is collapsing, the voltage at output is still holding on for now. From 3.5 to 5 amperes, the inductor declines in effectiveness until it collapses entirely.

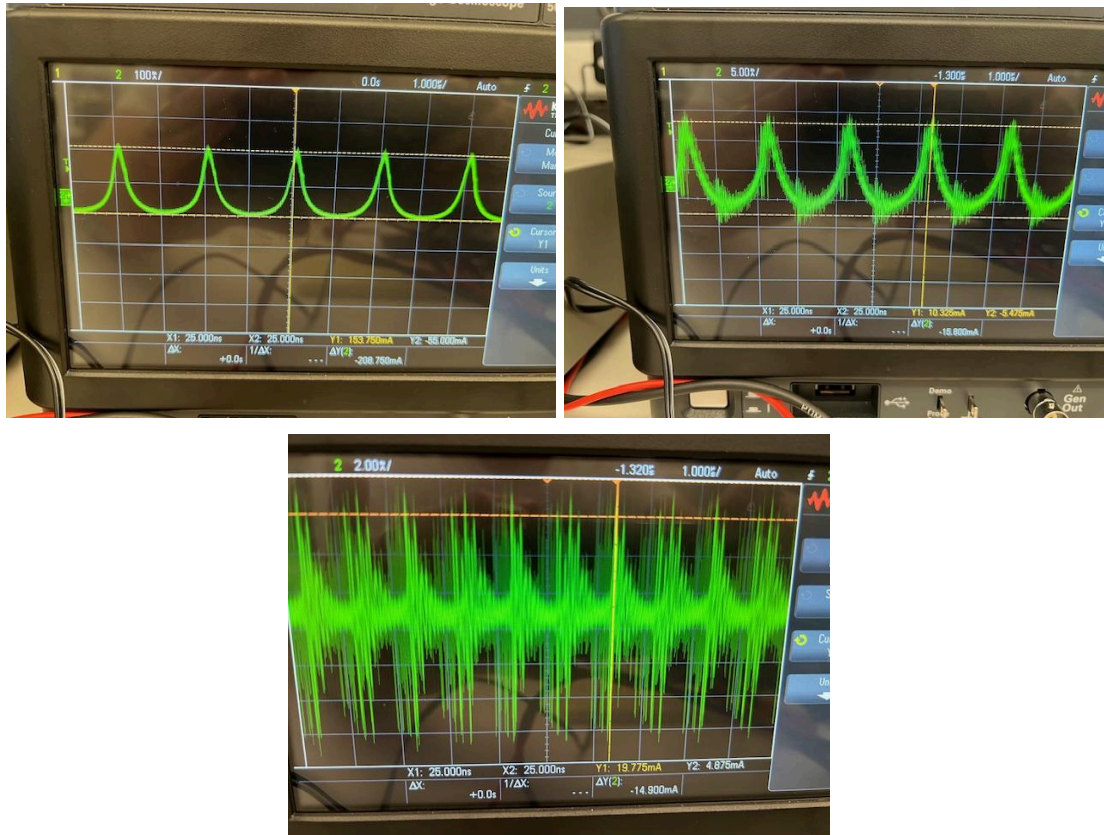


Figure 3-5: Inductor current ripple at 3.5, 4, and 5A respectively

At 5A, the output voltage begins to drop out. The inductor is no longer able to store energy. With no effective inductance, the current becomes noisy and less predictable since it can no longer generate voltage across it. The current oscillates wildly and the switching noise becomes the dominant driver in the circuit.

Discussion of Results:

This result was unexpected. The inductor had a 2.2uH value and a rated saturation current of 14A. The suspected cause of the disruption is inductor core material, and loss of induction because of high temperature.

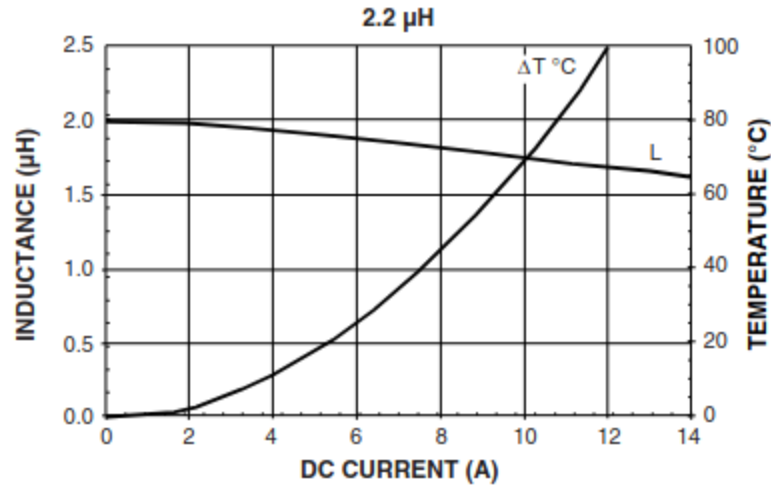


Figure 6: Characteristic Inductance, Temperature, Current chart (Vishay-Dale)

This chart shows 5 Amps reasonably within the operating range of the inductor. For this reason, it was chosen as a safe bet for this project. However, the temperature of the board switching at 500kHz is suspected of already being high, adding to the ΔT characteristic. It is inferred that saturation occurs sooner and temperature of the board is higher than what was assumed, leading to a drop off in inductive capability at load.

Although the datasheet does not specify a powdered core specifically, it does advertise soft saturation and excellent energy storage characteristics up to 5MHz, so it is unlikely that the core material is the cause of the problem. In further tests, it would be ideal to measure the temperature of the components while operating at full load to confirm or disconfirm this hypothesis.

Part 2: Output Capacitors

At the output are 3 Murata 10uF 10V rated capacitors with a total derated capacitance of $\approx 27\mu F$ and a total equivalent resistance of $\approx 1m\Omega$. Output voltage ripple was measured at the output using the oscilloscope and at with a 2A load.

This capacitance was selected using the equation:

$$C_{out} = \frac{1}{8 \times f_{sw} \times \left(\frac{\Delta V_{out}}{\Delta i} \right) - R_{esr}} \quad \text{eq. 1}$$

Which when carried out gave a C_{out} value of $\approx 22.7\mu F$. The 3 parallel $10\mu F$ capacitors were an ideal selection since they were within a few μF of my target, and definitely supplied enough capacitance to give me a small ΔV_{out} .

Once the capacitors, were selected and implemented the theoretical ΔV_{out} was calculated by rearranging eq. 1

$$\Delta V_{out} = \Delta i \times (R_{esr} + \frac{1}{8 \times f_{sw} \times C_{out}}) = 1.04A \times (.001 + \frac{1}{8 \times 500kHz \times 27\mu F}) = 0.0106V$$

This value falls well within the specified parameters, and the test results show that it was not exact.

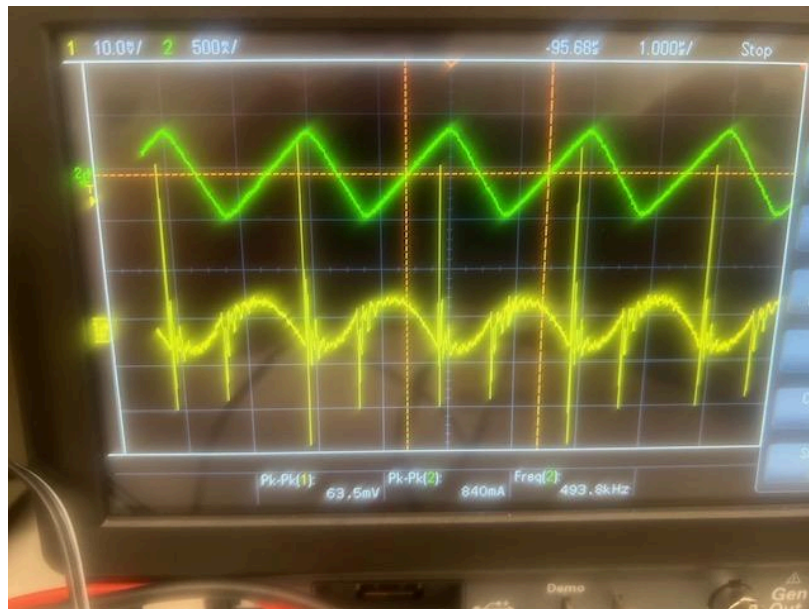


Figure 7: Output voltage ripple at 2A load

The test at output shows a characteristic voltage ripple with some large transient spikes at the switching frequency. The ripple itself falls within specifications at $\approx 15mV$, less that 1% of the 1.8V output. However, these transient spikes are showing up, and as we will see, will show up almost everywhere in the circuit.

Table 2: Peak to Peak ΔV

Theoretical (mV)	Actual (mV)
10.6	15

Analysis:

The peak to peak output voltage ripple falls within the specifications of the converter, not accounting for the transient spikes. Exploration of the source of voltage spikes will be conducted in subsequent sections of the report, however since they appear at the switching frequency, it is inferred to be connected to the switching IC transients being propagated through the circuit.

Part 3: Input Voltage Capacitors

The input voltage capacitors were chosen initially to have enough capacitance to manage inrush current. Ceramic capacitors were chosen for their low equivalent series resistance to prevent losses and overheating, while also having high enough admittance to allow current to be absorbed and protect the IC. Aiming for 1% peak to peak voltage ripple at input, the following equation is applied:

$$C_{in} = I_{out} \times (D \times (1 - D)) \times \frac{1}{f_{sw} \times \Delta V_{in}} = 5A \times (.36 \times .64) \times \frac{1}{500kHz \times 0.05} \approx 46\mu F$$

However, the actual capacitance value was overshoot, and three 100uF capacitors with a total of $\approx 150\mu F$ were installed at the input. It was assumed that more capacitance is good enough for the application, but results show an interesting phenomenon.

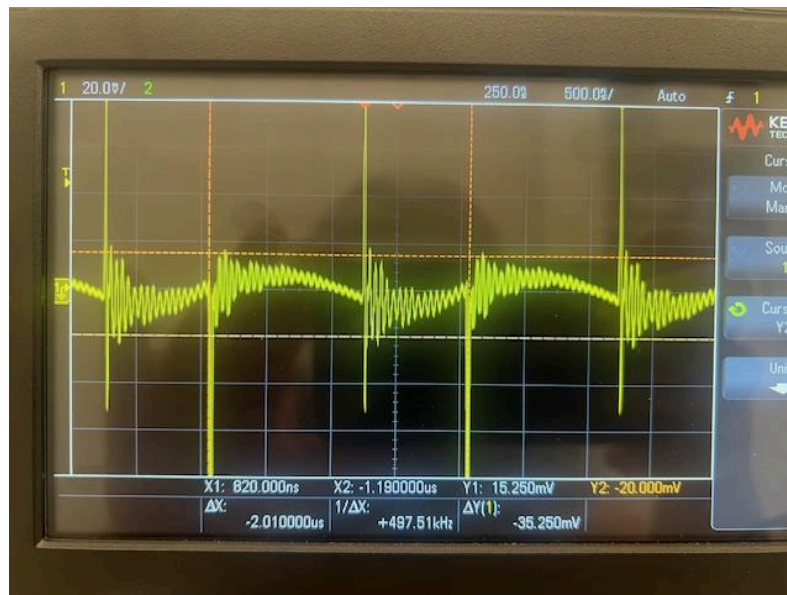


Figure 8: Input ripple at 150uF with ringing

As can be seen, the input voltage waveform shows typical peak to peak ripple voltage shape, but with a large transient spike and consistent ringing. It was hypothesized that the ringing could be explained perhaps by haphazard capacitor selection. It was decided to remove one capacitor at the input, lowering the

effective capacitance to 100uF, still double the value prescribed theoretically. I also added a 10uF capacitor in place of the other one, guessing that adding a lower value capacitor would help stabilize the ringing.

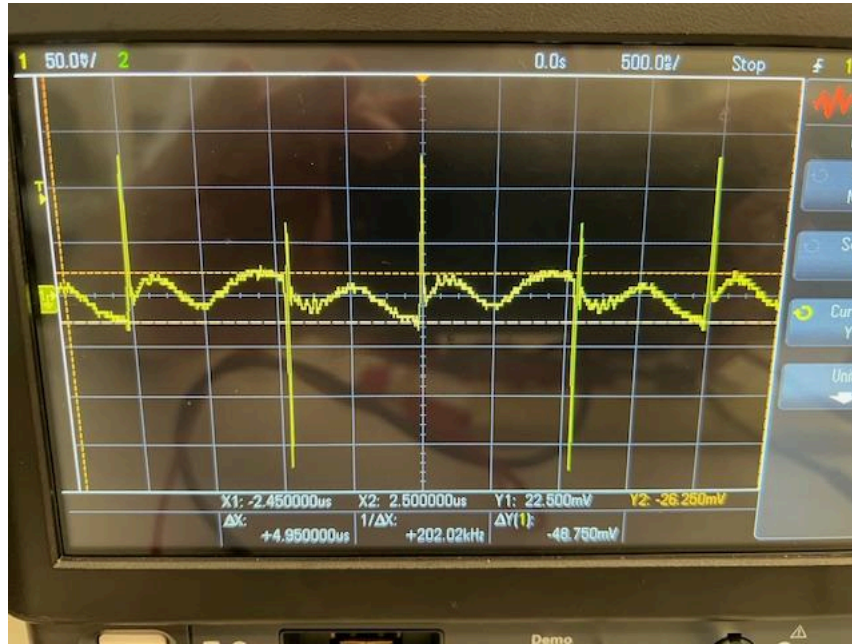


Figure 9: Input Voltage with 100uF effective capacitance

Lowering the input capacitance cleaned up the signal quite well. It can be seen that there are still voltage spikes, but further testing by lowering the capacitance further could begin to explain the unwanted transients. The peak to peak voltage ripple sits at about 50mV which is exactly the specification value designed for. Rearranging the C_{in} equation, we can see how closely our measurement aligns with the theoretical value.

$$\Delta V_{in} = I_{out} \times (D \times (D - 1)) \times \frac{1}{f_{sw} C_{in}} = 3A \times (.36 \times .64) \times \frac{1}{500kHz \times 100\mu F} = 0.014V$$

Table 3: Input ΔV

Theoretical (V)	Experimental(V)
0.014	0.049

Analysis:

Through experimentation with capacitor values, I was able to find strong evidence of a source for noise in the circuit. The addition of the small capacitor could be superfluous. Given more time, a regimen of testing input capacitance,

and noise at various locations in the circuit could prove to find a solution to this problem. Nevertheless, lowering the overall input capacitance improved the performance, and lowered noise in the circuit overall.

Part 4: Voltage Droop Transient

Calculating the compensation component values is imperative to achieving an acceptable voltage droop. Finding these values within the constraints of having already decided on the power path values proved challenging. A fuller picture of how these values are obtained is available in the appendix.

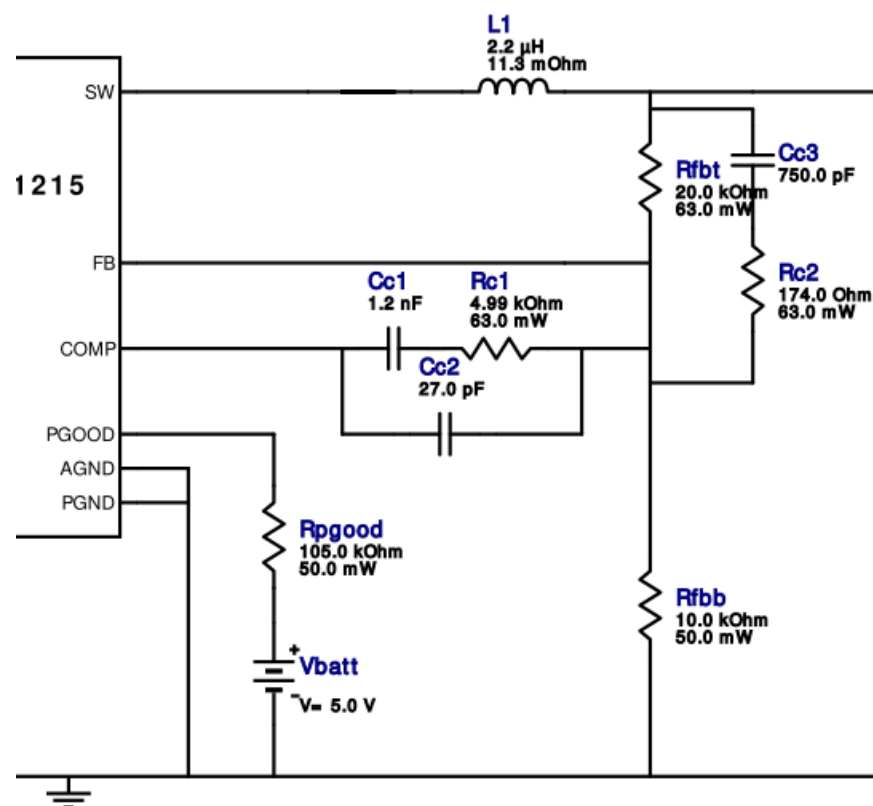


Figure 10: Compensation and Feedback loops

The values selected for the feedback and compensation loops were as close as possible to those seen in Figure 10. The results of a 2A to 3A load step can be viewed in Figure 11 and 12.

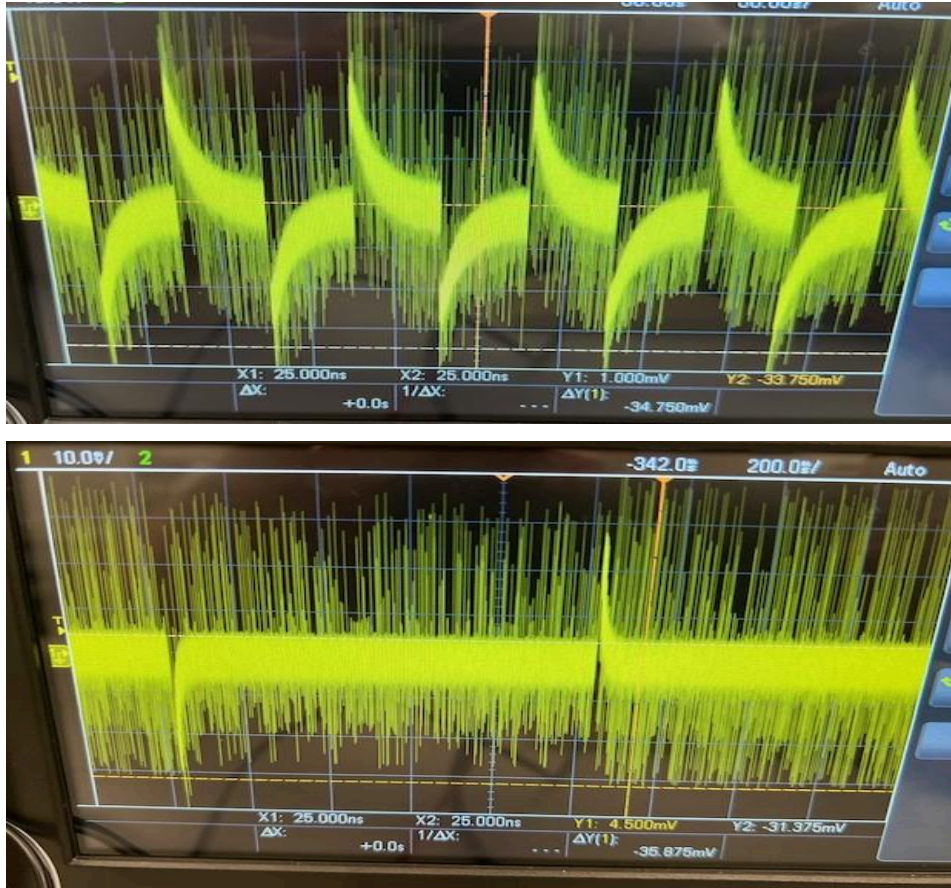


Figure 11 and Figure 12: Load step analysis

By applying a sudden load step at a 1Hz frequency, we are able to evaluate the buck converter's step response. The step is analyzed at the output voltage node and is characterized by the sudden spike in voltage. Ignoring the switching noise, we can see a stiff load step response, although the overall response is hard to gauge because of noise. We can discern from using the oscilloscope cursors (ignoring switching noise) a voltage droop of $\approx 35mV$ voltage droop at the load step.

Calculating V-Droop approximately

$$V_{droop} = \Delta i \times R_{esr} + \frac{L \times \Delta i_{step}}{C_{out} \times (V_{out} - V_{in})} = 1A \times 0.001 + \frac{2.2\mu H \times 1^2}{27\mu F (5V - 1.8V)} \approx 0.0265V$$

Table 4: V_{droop} values

Theoretical (mV)	Experimental (mV)
26.5	35.0

Part 5: Compensation Frequency Response and Phase Margin

To calculate the phase margin of our control loop, we rely on software rather than the limited instrumentation we have at our disposal. Ideally, a tool like LTSpice would be appropriate for this test, however, LTSpice simulations seemed unrepresentative and simulation times were very long. Instead, an online tool provided by Texas Instruments called WEBench was used to analyze various parameters of the designed circuit. The full WEBench report will be appended to this report.

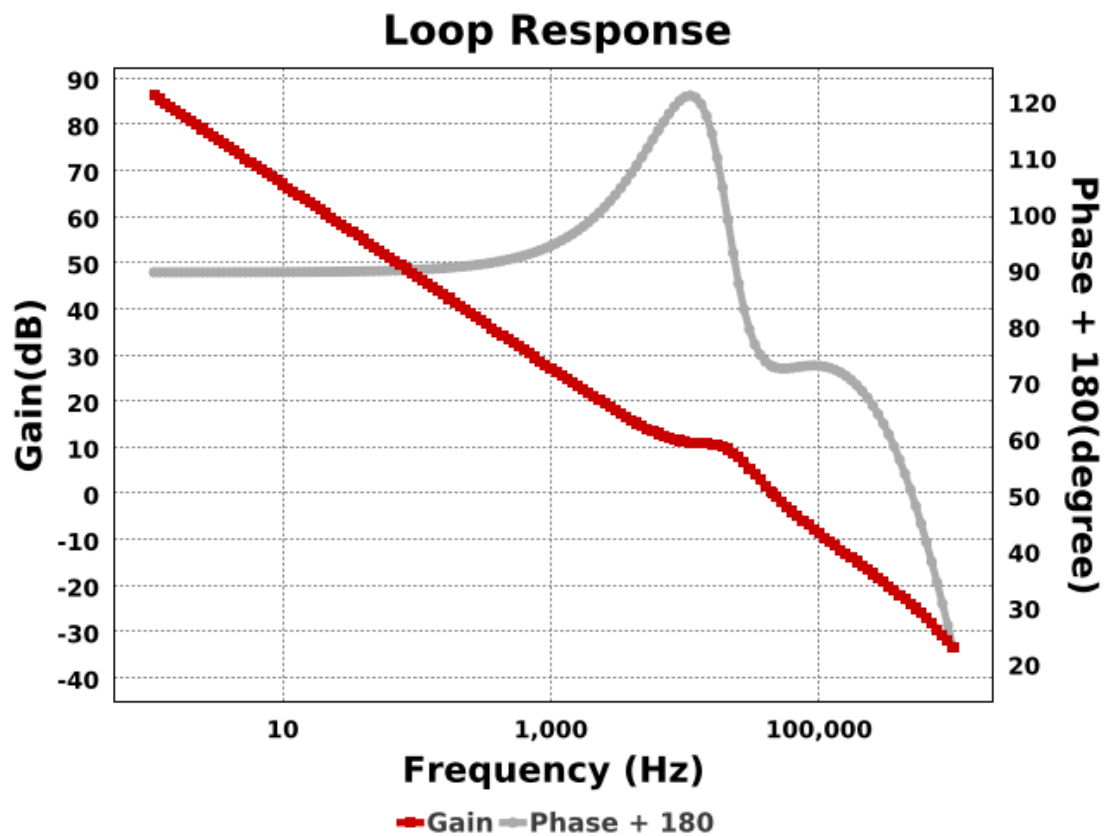


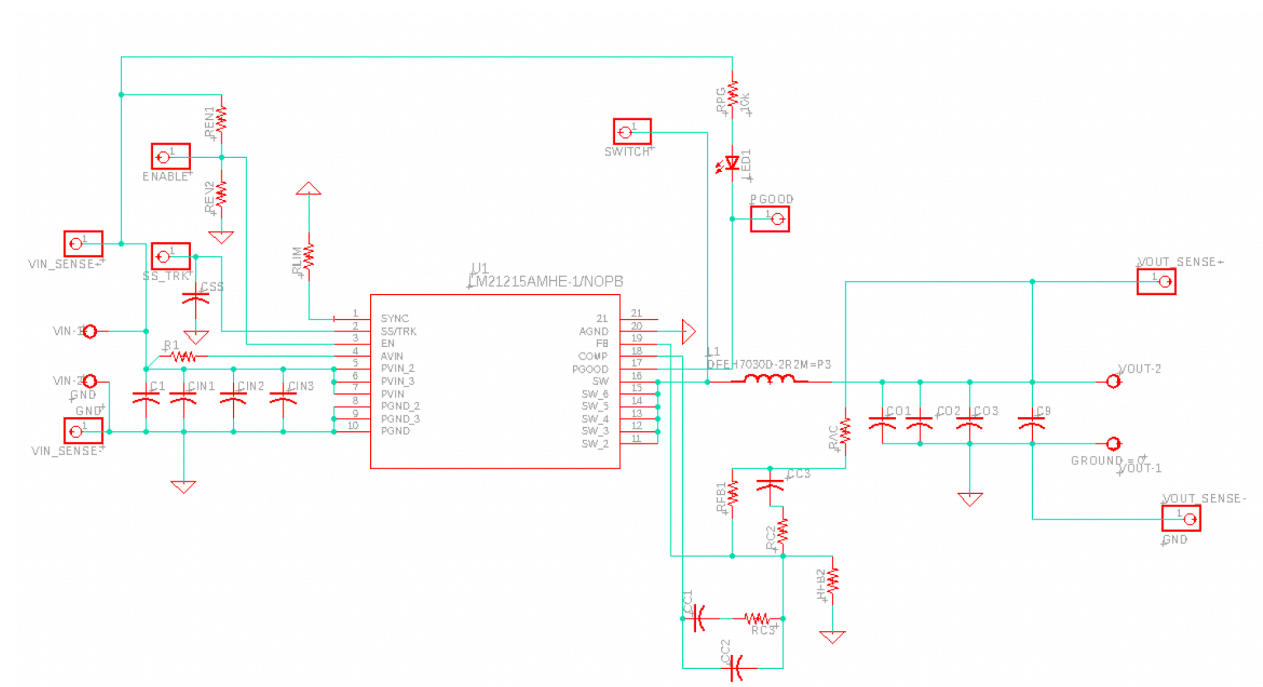
Figure 13: Bode plot of Circuit.

The goal of greater than 50 degrees phase margin is fairly well handled. By inspection of the Bode plot, the phase margin appears to be around 73 degrees, which is exactly what the appended simulation report shows. This phase margin gives plenty of room to further tune the other components, and improve the functionality of the circuit if we so wish.

Conclusions:

Although the circuit has obvious flaws, through testing we are able to narrow down the root cause of those flaws and plan for a better converter. Further testing is needed, as the efficiency of the circuit was not measured. The calculated efficiency from simulation of 90.6% is unlikely to be accurate, since at higher current loads, the saturation and suspected temperature of the inductor and the IC could raise the internal resistance and thus contribute to excessive power losses, especially at full load.

Circuit Schematic:



Board Layout:

